

Question Paper

Exam Date & Time: 20-Nov-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
SEVENTH SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

Generative AI and Large Language Models [NPT_4121]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Evaluate how the properties of word embeddings contribute to downstream NLP model performance. Compare traditional one-hot encoding with embedding-based representations and justify why embeddings outperform them in semantic understanding. (5)
- A)
- Course Outcome: "CO1" / Cognitive Level: "Analysis"
- B) Apply your understanding of the **NLP pipeline** to explain how raw text is transformed into model-ready input. Identify any three key stages and describe their purpose briefly. (3)
- Course Outcome: "CO2" / Cognitive Level: "Application"
- C) During the training of a Recurrent Neural Network (RNN) for predicting stock prices, the model struggles to learn long-term dependencies in the data. As an ML engineer, you are asked to suggest an appropriate training method to improve the model's learning process. Analyze the purpose of Backpropagation Through Time (BPTT) in training RNNs and demonstrate how it helps the model capture temporal dependencies in sequential data. (2)
- Course Outcome: "CO3" / Cognitive Level: "Application"
- 2) Assess how **Embeddings from Language Model (ELMo)** differ from static embeddings like Word2Vec or GloVe. Evaluate ELMo's advantages in representing contextual meaning and its impact on NLP model performance. (5)
- A)
- Course Outcome: "CO2" / Cognitive Level: "Analysis"
- B) Examine how **BERT's bidirectional attention** mechanism enables better understanding of context compared to unidirectional models like GPT. (3)
- Course Outcome: "CO3" / Cognitive Level: "Application"
- C) **Use** your understanding of **Large Language Models (LLMs)** to **illustrate** two major characteristics that **set them apart** from traditional NLP systems and **show** how these characteristics enhance real-world language processing tasks. (2)
- Course Outcome: "CO4" / Cognitive Level: "Application"
- 3) Critically analyze how identifying the **right use case** influences the success of a **Generative AI project**. Propose criteria for evaluating whether a use case is suitable for generative modeling. (5)
- A)
- Course Outcome: "CO3" / Cognitive Level: "Analysis"

- B) Apply the concept of **Representation Learning** to explain how deep neural networks automatically extract useful features from raw data, reducing manual feature engineering. (3)

Course Outcome: "CO4" / Cognitive Level: "Application"

- C) Apply your understanding of Foundation Models to identify three major models that have advanced the field of Artificial Intelligence and explain how each has contributed to practical AI innovations. (2)

Course Outcome: "CO5" / Cognitive Level: "Application"

- 4) **Design** an experiment using **OpenAI Playground** to evaluate the **emergent abilities** of a Transformer-based Large Language Model. Discuss how you would measure performance and interpret results. (5)
- A)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- B) Examine how **Prompt Engineering (PE)** enhances the output quality of Generative AI models. Explain using an example of rewriting prompts for better accuracy. (3)

Course Outcome: "CO5" / Cognitive Level: "Application"

- C) **Use** your understanding of **text generation techniques** to **differentiate** between **Top-k** and **Top-p (nucleus)** sampling and **illustrate** how each method influences the **diversity and coherence** of generated text. (2)

Course Outcome: "CO1" / Cognitive Level: "Application"

- 5) **Evaluate** the architectural design of **Generative Pre-trained Models (GPT-like)** and justify how pre-training and fine-tuning stages together enable state-of-the-art text generation capabilities. (5)

- A) Course Outcome: "CO5" / Cognitive Level: "Analysis"

- B) Demonstrate how **AI agents** that can use tools (e.g., calculators, search APIs, or code interpreters) enhance the performance of Large Language Models in real-world applications. Provide one example. (3)

Course Outcome: "CO1" / Cognitive Level: "Application"

- C) Apply your understanding of Natural Language Processing (NLP) to describe two fundamental goals of NLP in Artificial Intelligence and demonstrate how these goals are achieved through real-world applications. (2)

Course Outcome: "CO2" / Cognitive Level: "Application"

-----End-----

Event Name	Subject
------------	---------

058-END SEMSTER-ODD-2025 NPT_4121:Generative AI and Large Language Models

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	10
3	3	CO 3	10
4	4	CO 4	10
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
------------	---------

058-END SEMSTER-ODD-2025 NPT_4121:Generative AI and Large Language Models

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	10	25(50%)
4	Analysis	5	25(50%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
	Total	15	50 (100%)